Configuring PPPoE Dial-up for Internet Access

Difficulty (out of 3 stars): ★★★ Recommended study time: 2 hours

Contents

1 About This Lab

1.1 Task Description

Point-to-Point Protocol over Ethernet (PPPoE) provides point-to-point (P2P) connections on Ethernet networks and establishes PPP sessions to connect hosts on Ethernet networks to a remote broadband remote access server (BRAS). PPPoE features wide applications, high security, and convenient accounting.

After completing this task, you will gain a clear understanding of PPPoE and can configure PPPoE for Internet access.

1.2 Objectives

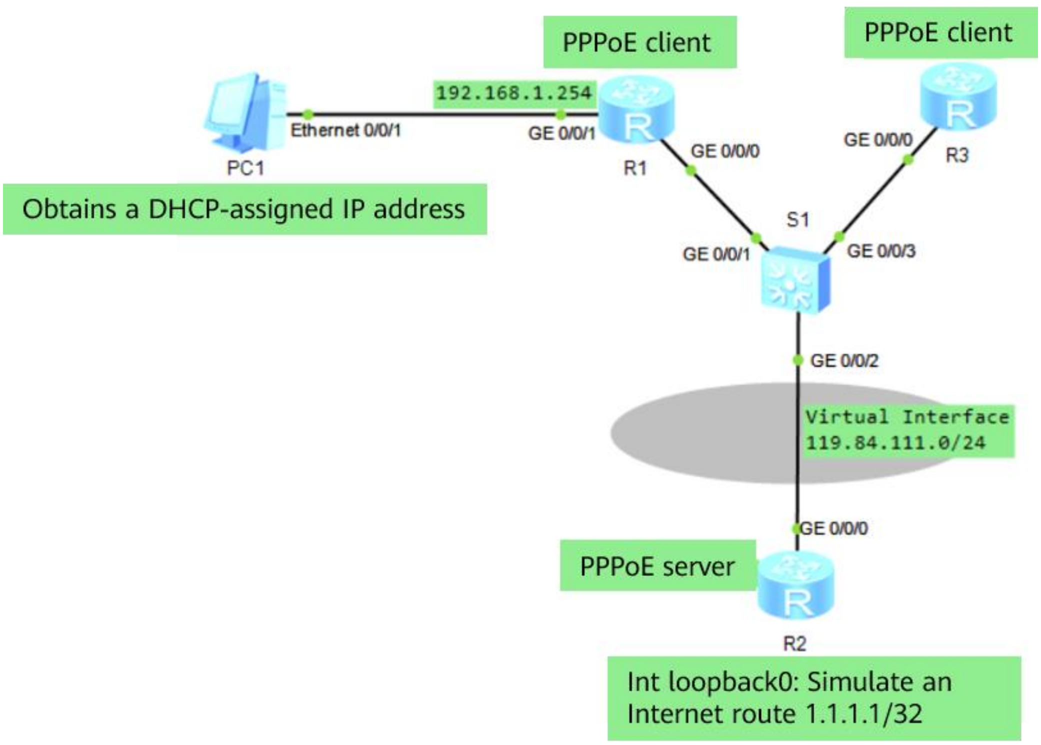
On completing this task, you will be able to:

1. Configure a dialer interface on a PPPoE client.
2. Configure PPPoE client authentication.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Lab topology for configuring PPPoE dial-up for Internet access



R1 and R3 are edge routers of an enterprise branch and connected to an ISP network through a PPPoE server (R2). You need to configure the PPPoE client on the edge routers of the enterprise so that hosts on the LAN can access Internet resources through PPPoE dial-up.

2.2 Initial Configuration

- Initial configuration of R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]ip address 192.168.1.254 24
[R1-GigabitEthernet0/0/1]quit
[R1]
```

- Initial configuration of R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]
[R2]int LoopBack 0            # Create an address for testing.
[R2-LoopBack0]ip address 1.1.1.1 32
[R2-LoopBack0]quit
[R2]
```

- Initial configuration of R3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]
```

- Initial configuration of PC1 (configuring PC1 to obtain an IP address through DHCP)

3 Task Implementation

3.1 Checking Basic Configurations

Omitted

3.2 Configuring the DHCP Server Function on R1

- Configure the DHCP server function on R1 to assign an IP address to the PC. An interface address pool is used.

```
[R1]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]dhcp select interface
[R1-GigabitEthernet0/0/1]dhcp server dns-list 114.114.114.114
[R1-GigabitEthernet0/0/1]quit
[R1]
```

- Check the IP address on the PC.

```
PC>ipconfig

Link local IPv6 address...........: fe80::5689:98ff:fe2e:555e
```

```
IPv6 address......................: :: / 128
IPv6 gateway.....................: ::
IPv4 address......................: 192.168.1.253
Subnet mask.......................: 255.255.255.0
Gateway...........................: 192.168.1.254
Physical address..................: 54-89-98-2E-55-5E
DNS server........................: 114.114.114.114

PC>ping 192.168.1.254

Ping 192.168.1.254: 32 data bytes, Press Ctrl_C to break
From 192.168.1.254: bytes=32 seq=1 ttl=255 time=31 ms
From 192.168.1.254: bytes=32 seq=2 ttl=255 time=31 ms
From 192.168.1.254: bytes=32 seq=3 ttl=255 time=16 ms
From 192.168.1.254: bytes=32 seq=4 ttl=255 time<1 ms
From 192.168.1.254: bytes=32 seq=5 ttl=255 time=15 ms

--- 192.168.1.254 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 0/18/31 ms

PC>
```

3.3 Configuring the PPPoE Server

1. Configure R2 as the PPPoE server.

   Although the PPPoE server is not on the enterprise network, you need to configure the PPPoE server in this lab to authenticate R1 and R3 on the enterprise network.

```
[R2]ip pool pool1
Info: It's successful to create an IP address pool.
[R2-ip-pool-pool1]network 119.84.111.0 mask 255.255.255.0
[R2-ip-pool-pool1]gateway-list 119.84.111.254
[R2-ip-pool-pool1]quit
[R2]
[R2]interface Virtual-Template 1
[R2-Virtual-Template1]ppp authentication-mode chap
[R2-Virtual-Template1]ip address 119.84.111.254 255.255.255.0
[R2-Virtual-Template1]remote address pool pool1
[R2-Virtual-Template1]quit
[R2]
```

2. Bind a virtual template to G0/0/0 of R2.

```
[R2]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]pppoe-server bind virtual-template 1
[R2-GigabitEthernet0/0/0]quit
[R2]
```

3. Configure valid accounts and passwords for the PPPoE clients.

```
[R2]aaa
[R2-aaa]local-user huawei1 password cipher huawei123
```

```
Info: Add a new user.
[R2-aaa]local-user huawei1 service-type ppp
[R2-aaa]
[R2-aaa]local-user huawei2 password cipher huawei123
Info: Add a new user.
[R2-aaa]local-user huawei2 service-type ppp
[R2-aaa]quit
[R2]
```

3.4 Configuring the PPPoE Client

1. Configure R1 as the PPPoE client.
   - Create a dialer interface on R1 and enable PPP authentication. Configure the user name and password for the PPP client (the user name and password must be the same as those on the PPPoE server).

```
[R1]dialer-rule
[R1-dialer-rule]dialer-rule 1 ip permit
[R1-dialer-rule]quit
[R1]
[R1]interface Dialer 1
[R1-Dialer1]dialer user user1
[R1-Dialer1]dialer-group 1
[R1-Dialer1]dialer bundle 1
[R1-Dialer1]ppp chap user huawei1
[R1-Dialer1]ppp chap password cipher huawei123
[R1-Dialer1]dialer timer idle 300
[R1-Dialer1]dialer queue-length 8
[R1-Dialer1]ip address ppp-negotiate
[R1-Dialer1]quit
[R1]
```

   - Bind the PPPoE dialer interface to the outbound interface.

```
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]pppoe-client dial-bundle-number 1
[R1-GigabitEthernet0/0/0]quit
[R1]
```

   - Configure a default static route from the PPPoE client to the PPPoE server.

```
[R1]ip route-static 0.0.0.0 0.0.0.0 Dialer 1
```

2. Configure R3 as the PPPoE client.

   Configure R3 as the PPPoE client. The configuration procedure is the same as that on R1.

```
[R3]dialer-rule
[R3-dialer-rule]dialer-rule 1 ip permit
[R3-dialer-rule]quit
[R3]
[R3]interface Dialer 1
[R3-Dialer1]dialer user user2
[R3-Dialer1]dialer-group 1
[R3-Dialer1]dialer bundle 1
[R3-Dialer1]ppp chap user huawei2
[R3-Dialer1]ppp chap password cipher huawei123
[R3-Dialer1]dialer timer idle 300
```

```
INFO:   The configuration will become effective after link reset.
[R3-Dialer1]dialer queue-length 8
[R3-Dialer1]ip address ppp-negotiate
[R3-Dialer1]quit
[R3]
[R3]interface GigabitEthernet 0/0/0
[R3-GigabitEthernet0/0/0]pppoe-client dial-bundle-number 1
[R3-GigabitEthernet0/0/0]quit
[R3]
[R3]ip route-static 0.0.0.0 0.0.0.0 Dialer 1
[R3]
```

3.5 Checking the PPPoE Status

1. Run the **display pppoe-server session all** command on R2 to check the PPPoE session status and configuration.

```
<R2>display pppoe-server session all
SID Intf                     State OIntf          RemMAC            LocMAC
1     Virtual-Template1:0    UP      GE0/0/0      00e0.fc67.3af9    00e0.fcdf.207b
2     Virtual-Template1:1    UP      GE0/0/0      00e0.fcd8.2cee    00e0.fcdf.207b
<R2>
```

   The preceding command output shows that the session status is normal.

2. Check dialer interface information on R1 and R3 and ensure that the dialer interfaces can obtain an IP address from the PPPoE server.

```
<R1>display ip interface brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
The number of interface that is UP in Physical is 3
The number of interface that is DOWN in Physical is 2
The number of interface that is UP in Protocol is 2
The number of interface that is DOWN in Protocol is 3

Interface                  IP Address/Mask       Physical    Protocol
Dialer1                    119.84.111.253/32      up          up(s)
GigabitEthernet0/0/0       unassigned            up          down
GigabitEthernet0/0/1       unassigned            down        down
GigabitEthernet0/0/2       unassigned            down        down
NULL0                      unassigned            up          up(s)
<R1>

[R3]display ip interface brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
The number of interface that is UP in Physical is 3
The number of interface that is DOWN in Physical is 2
The number of interface that is UP in Protocol is 2
The number of interface that is DOWN in Protocol is 3
```

```
Interface                         IP Address/Mask      Physical    Protocol
Dialer1                           119.84.111.252/32    up          up(s)
GigabitEthernet0/0/0              unassigned           up          down
GigabitEthernet0/0/1              unassigned           down        down
GigabitEthernet0/0/2              unassigned           down        down
NULL0                             unassigned           up          up(s)
[R3]
```

3. Check the connectivity between R1 and R3.

```
<R1>ping 119.84.111.252
  PING 119.84.111.252: 56  data bytes, press CTRL_C to break
    Reply from 119.84.111.252: bytes=56 Sequence=1 ttl=254 time=100 ms
    Reply from 119.84.111.252: bytes=56 Sequence=2 ttl=254 time=100 ms
    Reply from 119.84.111.252: bytes=56 Sequence=3 ttl=254 time=90 ms
    Reply from 119.84.111.252: bytes=56 Sequence=4 ttl=254 time=90 ms
    Reply from 119.84.111.252: bytes=56 Sequence=5 ttl=254 time=100 ms

  --- 119.84.111.252 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 90/96/100 ms

<R1>


[R3]ping 119.84.111.253
  PING 119.84.111.253: 56  data bytes, press CTRL_C to break
    Reply from 119.84.111.253: bytes=56 Sequence=1 ttl=254 time=130 ms
    Reply from 119.84.111.253: bytes=56 Sequence=2 ttl=254 time=90 ms
    Reply from 119.84.111.253: bytes=56 Sequence=3 ttl=254 time=90 ms
    Reply from 119.84.111.253: bytes=56 Sequence=4 ttl=254 time=80 ms
    Reply from 119.84.111.253: bytes=56 Sequence=5 ttl=254 time=90 ms

  --- 119.84.111.253 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 80/96/130 ms

[R3]
```

3.6 Configuring NAT on R1

Configure NAT on R1 so that the PC can access the Internet.

```
[R1]
[R1]acl 2000
[R1-acl-basic-2000]rule 1 permit
[R1-acl-basic-2000]quit
[R1]
```

```
[R1]interface    dialer1
[R1-Dialer1]nat outbound 2000
[R1-Dialer1]quit
[R1]
```

Run the **ping** command on the PC to test the connectivity with Loopback0 on R2.

```
PC>ping 1.1.1.1

Ping 1.1.1.1: 32 data bytes, Press Ctrl_C to break
From 1.1.1.1: bytes=32 seq=1 ttl=254 time=63 ms
From 1.1.1.1: bytes=32 seq=2 ttl=254 time=62 ms
From 1.1.1.1: bytes=32 seq=3 ttl=254 time=47 ms
From 1.1.1.1: bytes=32 seq=4 ttl=254 time=63 ms
From 1.1.1.1: bytes=32 seq=5 ttl=254 time=46 ms

--- 1.1.1.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 46/56/63 ms

PC>
```

4 Verification

Omitted

5 Quiz

1. Why does the MTU of PPPoE frames need to be reduced?

   By default, the Ethernet supports a maximum payload of 1500 bytes. The PPPoE header is 6 bytes, and the PPP protocol ID is 2 bytes. Therefore, the MTU of PPPoE frames cannot exceed 1492 bytes.
2. What is the function of the **dialer bundle** command during PPPoE configuration?

   The **dialer bundle** command specifies a dialer bundle for a dialer interface. The device then associates a physical interface with the dialer interface through the dialer bundle.
3. What phases does the PPPoE session establishment process include?

   Discovery, Session, and Terminate phases